

Homework 3 Solutions

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A study was conducted to investigate two treatments for patients suffering from multiple sclerosis. 150 suffers of the disease were recruited into the study, and 75 were randomized to receive azathioprine (AZ) alone (group 1), and 75 were randomized to receive azathioprine plus methylprednisommne (AZ+MP, group 2). For each participant, a measure of auto-immunity, azathioprine AFCR, was planned at clinic visits at baseline (time 0, at initiation of treatment) and at 3, 6, 9, 12, 15, and 18 months thereafter. Multiple sclerosis (MS) affects the immune system. Low values of AFCR (approaching 0) are evidence that immunity is improving, which is hopefully associated with a better prognosis for suffers of MS. Also recorded for each subject was age at entry into the study and an indicator of whether or not the subject had had previous treatment with either of the study agents (0=no, 1=yes). The average age of the men across both treatment groups was 50.45, with SD 6.69. The primary scientific aims of the study are to investigate whether (i) both treatments (AZ or AZ+MP) lower AFCR over the 18 month period and (ii) whether treatment with AZ+MP results in different immune system response than does AZ alone, and, if so, how it is different in terms of response over time. It was also suspected that a subject's age and prior history might be related to their AFCR level at baseline and to the rate at which AFCR changes during the 18 month period. The square root of AFCR is the response variable of interest (square roots were taken so that the AFCR observations better satisfy the assumption of normality).

1. (20 points) Discuss the issues of study design and how they would relate to your decision to use a linear mixed model or a pattern mixture model.

This study is a randomized longitudinal trial with repeated AFCR measurements on the same subjects over time. That makes the observations within a subject correlated, so a linear mixed model is a natural primary analysis tool. It allows us to model subject-specific trajectories over the 18-month follow-up and use all available measurements, even if some participants miss visits.

A pattern mixture model could be used since the data are designed balanced. However, there are 6 measurements, which is a lot for a pattern mixture model. That is, if we did a heterogeneous variance that would be 6 variance parameters. Further, if we did an unstructured correlation that would have 15 correlation parameters. This is a lot for only 140 subjects.

So, I would go with the linear mixed model.

2. (80 points) Regardless of your answer to (1), complete an analysis of this data using a linear mixed model. This analysis should yield the best possible answer to the study's scientific aims.

This analysis should include:

- a. Exploratory data analysis (e.g., figures) and discussion of findings.
- b. How do we test the hypothesis of interest? Will you include any confounders or effect modifiers?
- c. Model fit: compare different mean and random effect structures.
- d. Model diagnostics checks.

Part a

The first thing I'd like to do is to look at a plot of the data by group. I'm also going to change group to be a factor.

```
afcr <- read.csv("afcr.csv",header = TRUE, na.strings = "",
                 stringsAsFactors = FALSE)
afcr <- afcr %>% mutate( group = as.factor(group) )
str(afcr)
```

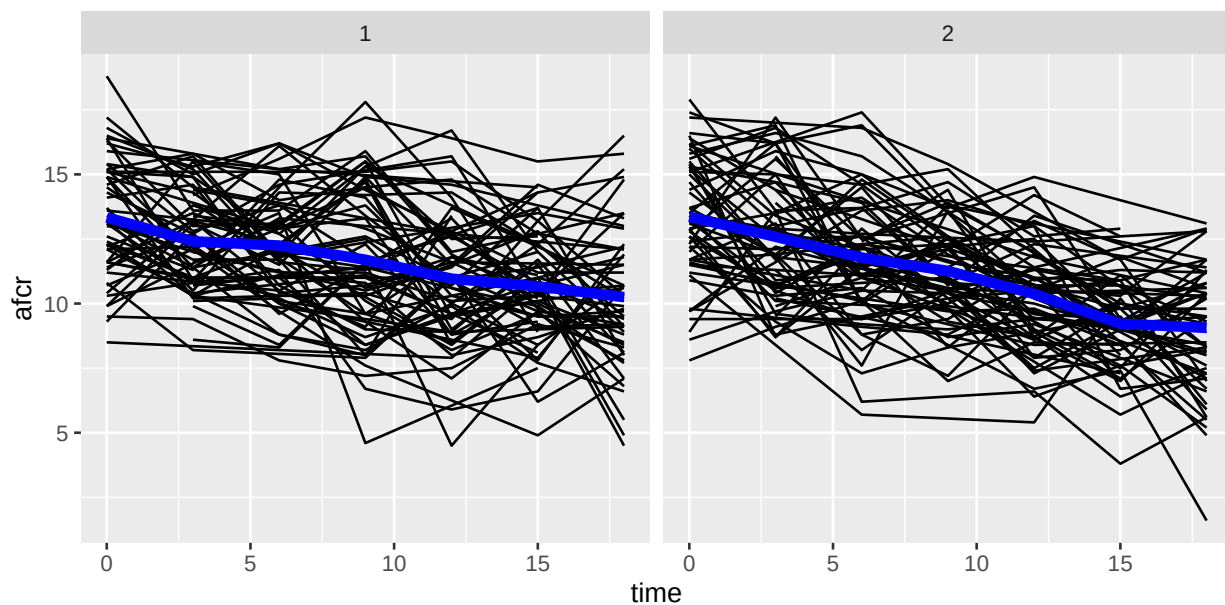
```
## 'data.frame': 790 obs. of 6 variables:
## $ id : int 1 1 1 1 2 2 2 2 2 3 ...
## $ time : int 0 12 15 18 0 6 9 12 18 0 ...
## $ afcr : num 8.5 7.9 10.7 11.5 11.2 10.5 9.1 13.7 10.7 11.4 ...
## $ group: Factor w/ 2 levels "1","2": 1 1 1 1 1 1 1 1 1 1 ...
## $ prior: int 0 0 0 0 0 0 0 0 0 0 ...
## $ age1 : int 47 47 47 47 48 48 48 48 48 53 ...
```

```
head(afcr)
```

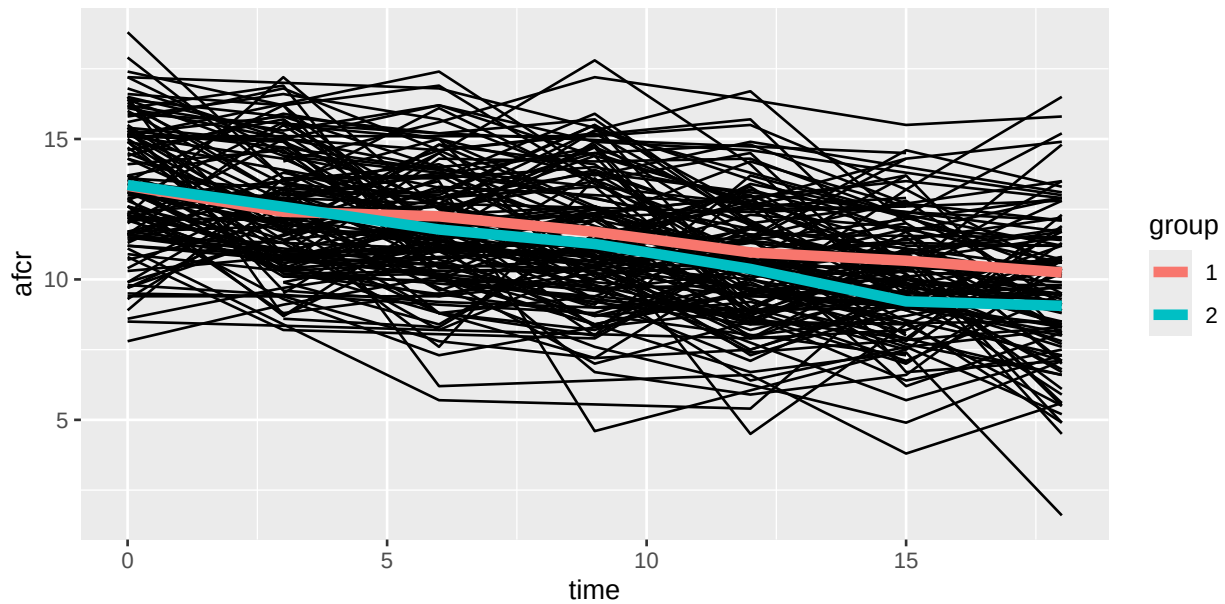
```
##   id time afcr group prior age1
## 1  1    0  8.5    1     0    47
## 2  1   12  7.9    1     0    47
## 3  1   15 10.7    1     0    47
## 4  1   18 11.5    1     0    47
## 5  2    0 11.2    1     0    48
## 6  2    6 10.5    1     0    48
```

```
p <- ggplot(data = afcr, aes(x = time, y = afcr, group = id))
```

```
# Separate plots for each group
p + geom_line() + stat_summary(aes(group = 1), geom = "line", fun = mean,
                              color = "blue", linewidth = 2) +
  facet_grid( . ~ group)
```



```
# Both groups on the same plot with means.
p + geom_line() + stat_summary(aes(group = group, color = group), geom = "line",
                              fun = mean, linewidth = 2)
```



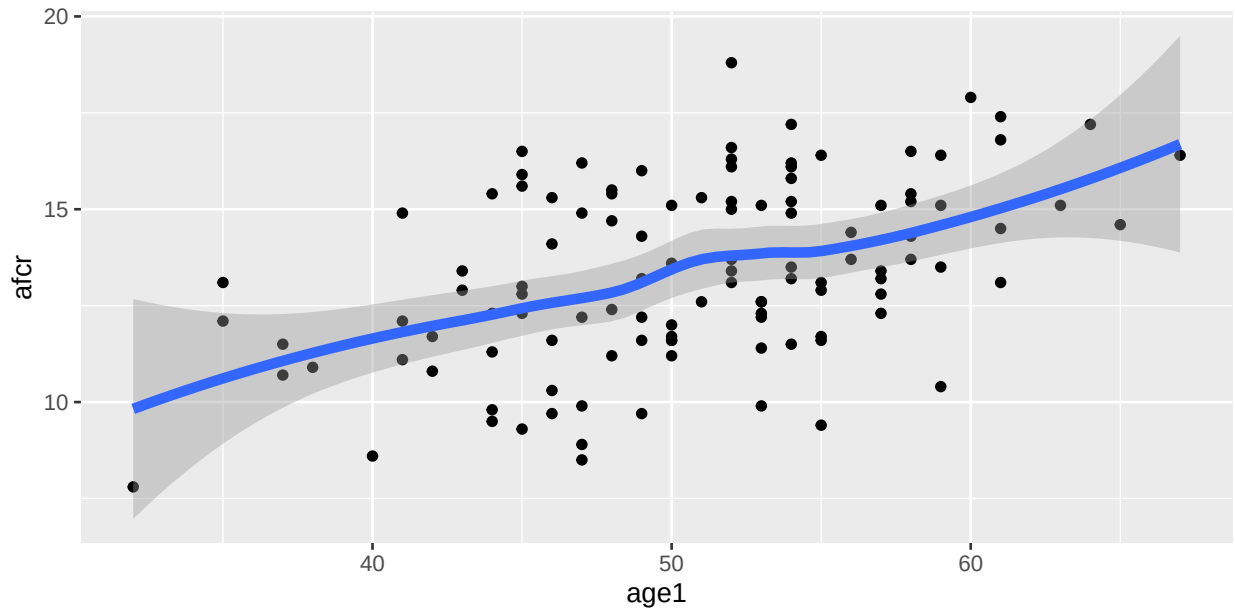
There doesn't appear to be much of an effect of the group variable. **It appears that a linear time trend may be sufficient for this data.** Also, there may be more heterogeneity in the later months of the study so a random slope may be required.

Checking baseline covariates

Now let's look at if age or prior have an impact on the baseline values of the outcome.

```
ggplot(data = afcr %>% filter(time == 0), aes(x = age1, y = afcr)) +
  geom_point() +
  geom_smooth(method = "loess", linewidth = 2)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



Clearly there is a relationship between age and afcr. The relationship appears to be pretty linear.

Now let's check the relationship between prior:

```
t.test(afcr ~ prior, data = afcr %>% filter(time == 0))
```

```
##
## Welch Two Sample t-test
##
## data: afcr by prior
## t = -2.1409, df = 104.8, p-value = 0.0346
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
## 95 percent confidence interval:
## -1.77299620 -0.06796918
## sample estimates:
## mean in group 0 mean in group 1
## 12.80638 13.72687
```

Again, a clear relationship here.

Let's see if there is a relationships between age, prior, and treatment.

```
t.test(age1 ~ group, data = afcr %>% filter(time == 0))
```

```
##
## Welch Two Sample t-test
##
## data: age1 by group
## t = -0.1061, df = 110.68, p-value = 0.9157
## alternative hypothesis: true difference in means between group 1 and group 2 is not equal to 0
## 95 percent confidence interval:
## -2.655869 2.385915
## sample estimates:
## mean in group 1 mean in group 2
## 50.50909 50.64407
```

```
chisq.test(afcr$group[afcr$time == 0], afcr$prior[afcr$time == 0])
```

```
##
## Pearson's Chi-squared test with Yates' continuity correction
##
## data: afcr$group[afcr$time == 0] and afcr$prior[afcr$time == 0]
## X-squared = 1.462, df = 1, p-value = 0.2266
```

No relationships. However, even in a randomized trial, it is reasonable to adjust for baseline covariates such as age and prior treatment when they are strongly associated with the outcome. Randomization ensures that treatment assignment is not confounded, so adjustment is not needed to remove bias. However, including prognostic covariates can improve precision by reducing residual variability and accounting for any chance imbalances between groups. In this setting, since age and prior treatment are clearly related to AFCR, adjusting for them will lead to more efficient estimates of the treatment effects without changing their interpretation.

Part b

The main scientific question is whether the two treatment groups differ in how AFCR changes over time. The primary test is therefore the treatment $\times$ time interaction. The model should have time, treatment, and treatment $\times$ time.

We don't need to adjust for confounders because the study is randomized. So age and prior treatment are not needed to control confounding of treatment assignment. However, they are reasonable to include as baseline prognostic covariates because they may improve precision and account for chance imbalance.

If we thought there were some effect modifiers, we would put them in using interaction terms. Here, if there is scientific reason to believe age or prior affect the rate of change over time, then they can also be included as effect modifiers through interactions with time, treatment, or treatment $\times$ time.

Part c

First, let's check for if a random slope is needed:

```
# Random intercept only
LMM_formula <- afcr ~ age1 + prior + group + time + group*time + (1|id)
LMM_linear_int <- lmer( formula = LMM_formula , data = afcr)

# Random intercept and slope only
LMM_formula <- afcr ~ age1 + prior + group + time + group*time + (1 + time|id)
LMM_linear_int_slp <- lmer( formula = LMM_formula , data = afcr)
```

```
## Warning in checkConv(attr("opt", "derivs"), opt$par, ctrl = control$checkConv, :
## Model failed to converge with max|grad| = 0.00460632 (tol = 0.002, component 1)
```

```
anova(LMM_linear_int, LMM_linear_int_slp)
```

```
## refitting model(s) with ML (instead of REML)
## Data: afcr
## Models:
## LMM_linear_int: LMM_formula
## LMM_linear_int_slp: LMM_formula
##
```

	npar	AIC	BIC	logLik	-2*log(L)	Chisq	Df	Pr(>Chisq)
## LMM_linear_int	8	3273.1	3310.5	-1628.5	3257.1			
## LMM_linear_int_slp	10	3276.8	3323.5	-1628.4	3256.8	0.3278	2	0.8488

Based on this analysis, a random slope is not required. So, we'll use the model with just the random intercept.

While the linear trend in afcr by time was apparent, we'll check a profile analysis just to be complete:

```
# Random intercept only (with profile)
LMM_formula <- afcr ~ age1 + prior + group + as.factor(time) + group*as.factor(time) + (1|id)
LMM_profile_int <- lmer( formula = LMM_formula , data = afcr)

anova(LMM_linear_int,LMM_profile_int)
```

```
## refitting model(s) with ML (instead of REML)

## Data: afcr
## Models:
## LMM_linear_int: LMM_formula
## LMM_profile_int: LMM_formula
##           npar    AIC    BIC logLik -2*log(L)  Chisq Df Pr(>Chisq)
## LMM_linear_int      8 3273.1 3310.5 -1628.5    3257.1
## LMM_profile_int    18 3287.9 3372.0 -1626.0    3251.9 5.1557 10      0.8805
```

So, the linear model looks to be the best.

Checking for effect modification (optional)

Now, let's check if effect modification by age or prior is present. First, I'll check this out for age. I'm only going to consider a three-way age1 x group x time interaction since that is the only one that would substantially change our conclusions.

```
# Random intercept with age effect modification
LMM_formula <- afcr ~ age1 + prior + group + time + group*time +
  age1*time + age1*group + age1*group*time + (1|id)
LMM_linear_int_age_int <- lmer( formula = LMM_formula , data = afcr)

anova(LMM_linear_int,LMM_linear_int_age_int)

## refitting model(s) with ML (instead of REML)

## Data: afcr
## Models:
## LMM_linear_int: LMM_formula
## LMM_linear_int_age_int: LMM_formula
##           npar    AIC    BIC logLik -2*log(L)  Chisq Df
## LMM_linear_int      8 3273.1 3310.5 -1628.5    3257.1
## LMM_linear_int_age_int 11 3277.4 3328.8 -1627.7    3255.4 1.7117 3
##           Pr(>Chisq)
## LMM_linear_int
## LMM_linear_int_age_int    0.6343
```

No effect modification for age. Now, let's check for prior:

```
# Random intercept with age effect modification
LMM_formula <- afcr ~ age1 + prior + group + time + group*time +
  prior*time + prior*group + prior*group*time + (1|id)
LMM_linear_int_prior_int <- lmer( formula = LMM_formula , data = afcr)

anova(LMM_linear_int,LMM_linear_int_prior_int)
```

```
## refitting model(s) with ML (instead of REML)
```

```
## Data: afcr
## Models:
## LMM_linear_int: LMM_formula
## LMM_linear_int_prior_int: LMM_formula
##               npar      AIC      BIC  logLik -2*log(L)  Chisq Df
## LMM_linear_int           8 3273.1 3310.5 -1628.5    3257.1
## LMM_linear_int_prior_int 11 3274.7 3326.1 -1626.3    3252.7 4.3935  3
##               Pr(>Chisq)
## LMM_linear_int
## LMM_linear_int_prior_int      0.222
```

No effect modification for prior So, we'll go back to the first model we fitted.

Part D

Here's a summary of the final model:

```
summary(LMM_linear_int)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: LMM_formula
## Data: afcr
##
## REML criterion at convergence: 3282.3
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.6963 -0.6739 -0.0090  0.6983  2.6068
##
## Random effects:
##  Groups   Name                Variance Std.Dev.
##  id       (Intercept)  0.8618    0.9283
##  Residual                  3.0838    1.7561
## Number of obs: 790, groups: id, 140
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   4.29370    0.77957 139.44563   5.508 1.70e-07 ***
## age1           0.16202    0.01491 132.12006  10.863 < 2e-16 ***
## prior          0.95689    0.20612 132.77669   4.642 8.17e-06 ***
## group2         0.32659    0.27727 399.30245   1.178    0.24
## time          -0.16083    0.01497 661.68769 -10.747 < 2e-16 ***
## group2:time    -0.08645    0.02112 660.50884  -4.094 4.77e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) age1   prior  group2 time
## age1          -0.953
## prior         -0.088 -0.084
## group2        -0.189 -0.002  0.073
## time          -0.164 -0.014  0.023  0.491
## group2:time    0.113  0.016 -0.027 -0.685 -0.709
```

Despite the first figure appearing that there was not an impact of the drug, the final model does show that the treatment was effective. Specifically, the $AZ + MP$ group had their azathioprine decrease more rapidly than the group that had AZ alone. In the AZ group we expect that the outcome ($\sqrt{afcr}$) will decrease 0.158 units for every month increase. In the $AZ + MP$ group, the outcome will decrease 0.089 units more than in the AZ group.